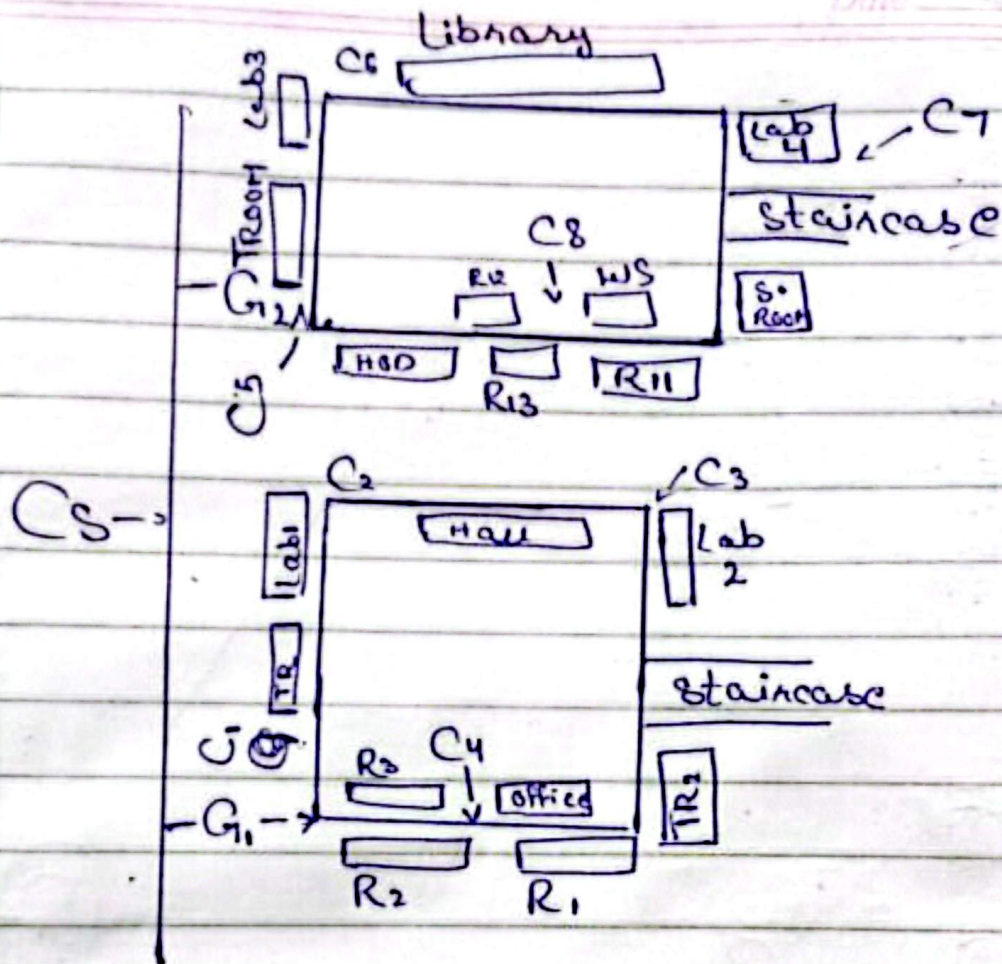


H.W

Date



Problem Lab 3 to Lab 4

BFS (Breadth First Search)

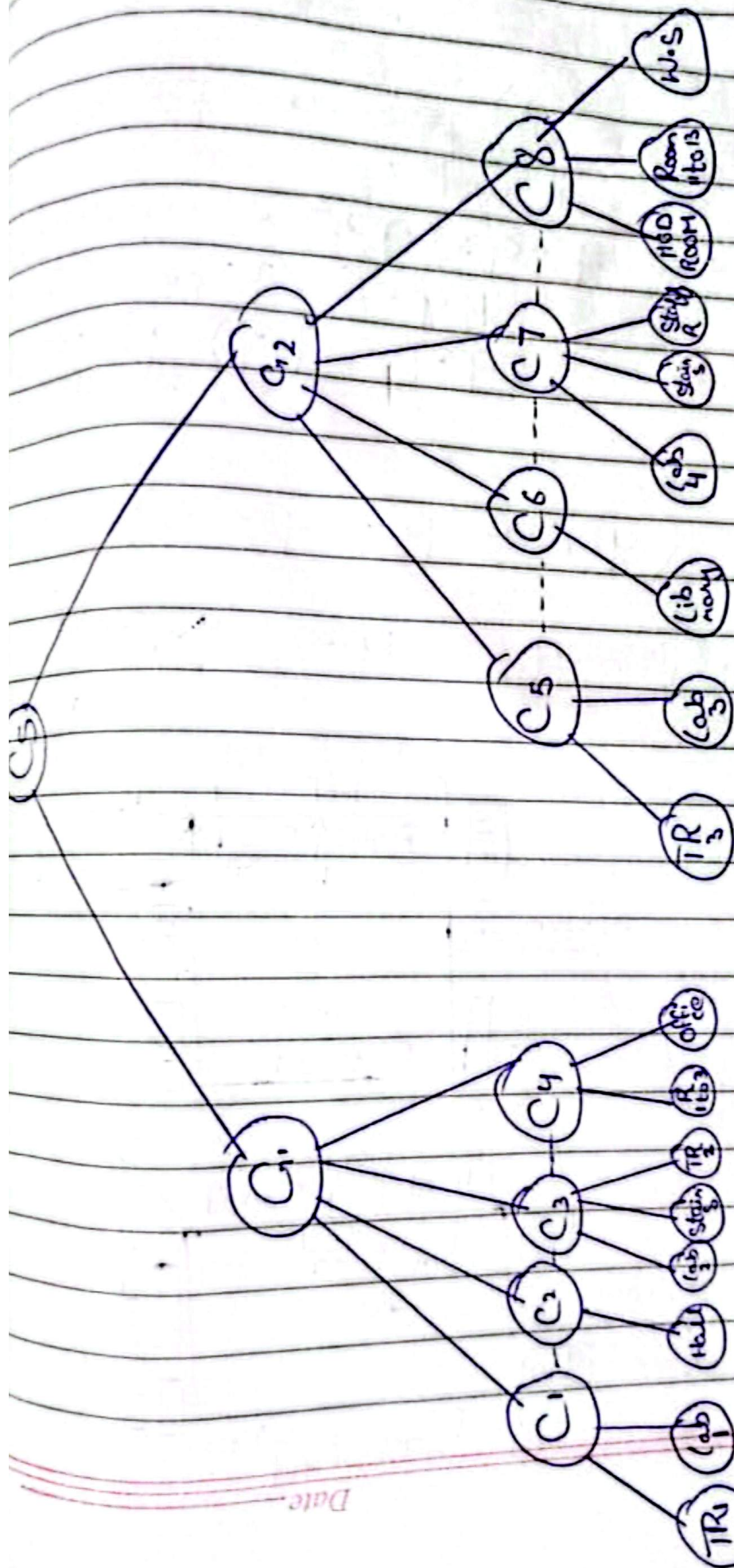
Pre Req:

- 1) Initial State = Lab 3
- 2) Goal State = Lab 4
- 3) Transition model = Lab 3 to connected places.
- 4) Path cost = no cost (give us shortest path)

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Table

$C_8 \rightarrow$	G_{11}, C_{12}
$G_1 \rightarrow$	C_1, C_2, C_3, C_4
$G_2 \rightarrow$	C_5, C_6, C_7, C_8
$C_1 \rightarrow$	TR_1, lab_1, C_2
$C_2 \rightarrow$	Hall, C_3
$C_3 \rightarrow$	Stairs, lab2, TR_2, C_4
$C_4 \rightarrow$	Office, TR_2 , Rooms (1 to 3)
$C_5 \rightarrow$	TR_3 , lab3, C_6
$C_6 \rightarrow$	Library, C_7
$C_7 \rightarrow$	lab4, Stairs, staff room, C_8
$C_8 \rightarrow$	HOD Room, R(11 to 13), W.S
lab1 $\rightarrow$	C_1

Hall $\rightarrow C_2$	HOD Room $\rightarrow C_8$
lab2 $\rightarrow C_3$	Room(11 to 13) $\rightarrow C_8$
Stairs $\rightarrow C_3$	W.S $\rightarrow C_8$
$TR_2 \rightarrow C_3$	
R, to 3 $\rightarrow C_4$	
Office $\rightarrow C_4$	
$TR_3 \rightarrow C_5$	
lab3 $\rightarrow C_5$	
Library $\rightarrow C_6$	
lab4 $\rightarrow C_7$	
Stairs2 $\rightarrow C_7$	
Staff room $\rightarrow C_7$	

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Q) Find lab3 to lab4
initialization

Q: [lab3]

V: [lab3]

Dequeue lab3 & expand lab3

Q: [C5]

V: [lab3, C5]

Dequeue C5 & expand it

Q: [TR3, C6, G12]

V: [lab3, C5, TR3, C6, G12]

Dequeue TR3 and expand it

Q: [C6, G12]

V: [lab3, C5, TR3, C6, G12]

Dequeue C6 and expand it

Q: [G12, G1, Lib]

V: [lab3, C5, TR3, C6, G12, G1, Lib]

Deq G12 & expand it.

Q: [G1, Lib, C8, C8]

V: [lab3, C5, TR3, C6, G12, G1, Lib, C8, C8]

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Req C₁ & expand it.

Q: [Lib, C₅, C₈, Lab₄, Stairs, Staff, R]
V: [Lab₃, C₅, TR₃, C₆, G₂, C₇, Lib, C₈, C₅, Lab₄,
Stairs, Staff, R]

Req Lib & expand it.

Q: [C₅, C₈, Lab₄, Stairs, Staff, R]
V: [Lab₃, C₅, TR₃, C₆, G₂, C₇, Lib, C₈, C₅, Lab₄,
Stairs, Staff, R].

Req C₅ & expand it

Q: [C₈, Lab₄, Stairs, Staff, R, G₁]
V: [Lab₃, C₅, TR₃, C₆, G₂, C₇, Lib, C₈, C₅, Lab₄,
Stairs, Staff, R, G₁]

Req C₈ and expand it

Q: [Lab₄, St, St.R, G₁, HOD ROOM, R, W.S]
V: [Lab₃, C₅, TR₃, C₆, G₂, C₇, Lib, C₈, C₅, Lab₄,
Stairs, St.R, Room 1103, G₁, HOD ROOM, W.S].

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